

SUHIL JUGROOP

Full-Stack Developer & AI Engineer

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EDUCATION

Bachelor of IT — Computational Science

Belgium Campus ITVersity
Graduated Cum Laude May 2026
73.5% Avg · 26 Distinctions
Top 9 in cohort

Key Modules:

- Programming 85%
- Web Programming 84%
- Business Intelligence 85%
- Data Analytics 82%
- Software Engineering 82%
- Cloud-Native Arch. 81%

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C#, Java
Frontend: React, Next.js 15, Three.js, R3F, Tailwind, Shadcn UI
Backend: Node.js, Express.js, REST APIs, Serverless, Firebase, Firestore, MongoDB, MySQL
AI & ML: Genkit, Gemini, YOLO, Neural Networks (NumPy), Scikit-learn, Pandas, Matplotlib, SciPy
Cloud & DevOps: Google Cloud, Azure, Docker, PWA, Git
Testing & Design: Cypress, Jest, UML, ERD, SDLC, Figma

LEADERSHIP

Led the 11-person Invascan capstone team (incl. 4 senior industry professionals) as project lead — managing sprint delivery, stakeholder reporting to lecturers, deans and external partners, and cross-discipline collaboration from requirements through final evaluation.

CONTACT

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PROFESSIONAL SUMMARY

Full-stack developer and recent IT graduate specialising in generative AI, cloud-native architecture, and production web systems. Graduated with a 73.5% weighted average and 26 distinctions from Belgium Campus ITVersity (top 9 in cohort). Delivered real-world client tools — including a live legal case management PWA and a production business website — alongside AI-integrated academic platforms using Next.js, TypeScript, Firebase, and Google AI (Genkit/Gemini). Experienced in system architecture, OOP design, data science, and leading cross-functional teams.

PROJECTS

LexDesk (Production Internal Tool) — Cloud-first PWA case management platform for legal practitioners

- Built complete platform: case files, financial dashboards, collection rate tracking
- Automated PDF generation for invoices, court pleadings, and correspondence via jsPDF
- Offline-first via Firestore, PWA install support, integrated calendar with .ics export, RBAC

Tech: Next.js 15 · Firebase · Firestore · Shadcn UI · Tailwind · jsPDF · PWA

CampusLearn — AI-enabled role-based academic platform (serverless Google Cloud)

- Multi-role system (students/tutors/lecturers/admins) with full auth, RBAC, course management, grading
- AI tutor using Gemini via Genkit with document retrieval; real-time Firestore sync; Cypress E2E coverage

Tech: Next.js · TypeScript · Firebase · Genkit · Gemini · Tailwind · Cypress

ScholarAI — GenAI personal tutor with adaptive learning

- Socratic dialogue, multi-modal inputs (text, voice, docs), adaptive quizzes, performance-based coaching

Tech: Next.js 15 · TypeScript · Genkit · Gemini · Zustand · Firebase

EduSphere Solutions (edu-sphere.co.za) — Production client website

- Designed and delivered a fully responsive production site for an IR consultancy, live and serving real clients

Tech: Next.js · Tailwind CSS · Shadcn UI

Invascan (Final Year Capstone — Team Lead) — AI-powered invasive species detection

- Led cross-functional team of 11 (incl. 4 industry professionals) across full project lifecycle
- Architected YOLO-based computer vision pipeline for real-time *Pyracantha* species detection
- Owned end-to-end ML pipeline: dataset acquisition, cleaning, preprocessing, YOLO config, training, tuning
- Built reporting API and data filtering logic to surface meaningful detection results
- Containerised entire system with Docker — multi-service orchestration for consistent dev/deployment
- Coordinated stakeholder reporting across lecturers, deans, and external partners

Tech: Python · Ultralytics YOLOv8 · Docker · Computer Vision · OpenCV · NumPy

MagCourt — Enterprise system architecture (Software Analysis & Design)

- Full SDLC: requirements engineering to multi-layer architecture to OOP class diagrams to physical ERDs to UI workflows to structured test plan (Unit/Integration/System/Acceptance)

Tech: UML · ERD · OOP Design · SDLC · Azure/Cloud Evaluation

COMPETENCIES

Data Science & ML: End-to-end pipelines (Pandas/NumPy) · EDA · Supervised/unsupervised ML · ANNs from scratch · Hypothesis testing · Bias-variance trade-offs

OOP Development: C# Windows Forms CRUD · Layered architecture · File persistence · Exception handling · Java servlets + DB

System Architecture: Multi-layer enterprise design · SDLC documentation · Build/buy/outsource evaluation